Exercise Sheet 2

Even though this PVK is structured into thematic sections and some content has not yet been covered, it is explicitly permitted and *should* be the case that in the problem sets *all the material* covered in the semester and in the lecture may be used.

Exercise 1
Calculate the value of the series

$$\sum_{n=1}^{\infty} \frac{\log(2^n)}{2^n}.$$

Exercise 2 (HS2022 G. Felder)
Calculate the value of the series

$$\sum_{n=1}^{\infty} \frac{1}{n2^n}.$$

Hint: You can differentiate the series to derive a general formula for $\sum_{n=1}^{\infty} \frac{1}{n} q^n$. Note that there are multiple ways of solving this exercise!

Exercise 3 (HS2022 G. Felder)
Calculate the radius of convergence of the power series

$$\sum_{n=0}^{\infty} \frac{(3n)!}{(n!)^3} x^n.$$

Exercise 4 (FS2024, A. Figalli)
Calculate the radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{n6^n}{3^{2n+1} + n2^n} x^n.$$

Exercise 5 (FS2025, L. Kobel-Keller)
The series

$$\sum_{n=1}^{\infty} \frac{n^n (x-2)^n}{(2n)!}$$

converges...

(A) only for $x = 0$.

(B) for all $x \in \mathbb{R}$.

(C) only for $x \in (-1, 1)$.

(D) only for $x \in (0, 4)$.

Exercise 6 (HS2022 G. Felder)
Calculate the maximum $m = \max\{3x^3 - x \mid x \in [-1, 1]\}$.

Exercise 7 (FS2023 G. Felder)
Calculate the minimum $m = \min\{x \log(x) \mid x > 0\}$.

Exercise 8
Let $D, E \subset \mathbb{R}$ be subsets and $x_0 \in D$ an accumulation point of $D\backslash\{x_0\}$. Let $f : D \to E$ be differentiable at x_0, such that $y_0 = f(x_0)$ is an accumulation point of $E\backslash\{y_0\}$ and $g : E \to \mathbb{R}$ differentiable at y_0. Show that $g \circ f : D \to \mathbb{R}$ is differentiable at x_0 and we have

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

Exercise 9
Let $D, E \subseteq \mathbb{R}$ and let $f : D \to E$ be a continuous, bijective function whose inverse $f^{-1} : E \to D$ is also continuous. Let $\bar{x} \in D$ be an accumulation point of $D \setminus \{\bar{x}\}$ and let f be differentiable at $\bar{x}$ with $f'(\bar{x}) \neq 0$. Prove, that f^{-1} is differentiable at $\bar{y} := f(\bar{x})$ and we have

$$(f^{-1})'(\bar{y}) = \frac{1}{f'(\bar{x})}.$$

Exercise 10
Let $f : [a, b] \to \mathbb{R}$ be a continuous function and differentiable on (a, b). Furthermore, assume $f(a) < 0 < f(b)$. Show that if $f'(x) \neq 0$ for all $x \in [a, b]$, then f has exactly one zero.